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19 January 1979
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MILITARY SPECIFICATION

FUEL SYSTEMS, AMPHIBIOUS ASSAULT AND TACTICAL AIRFIELD DISPENSING: GENERAL SPECIFICATION FOR

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers assemblies and components of 4- and 6-inch portable amphibious assault fuel systems and tactical airfield fuel dispensing systems.

1.2 Classification. The fuel systems shall be of the following types as specified in 3.3, 6.2.1, and 6.9:

Type I.	- Amphibious Assault Fuel System (AAFS).
Type II.	- Tactical Airfield Fuel Dispensing System (TAFDS).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4930

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SPECIFICATIONS

FEDERAL

W-C-440	- Clips, Electrical, General Specification for.
W-R-550	- Rod, Ground (With Attachments).
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

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MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture- and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-236	- Pipe, Culvert, Nestable.
MIL-H-370	- Hoses and Hose Assemblies, Rubber, Liquid Fuel, Noncollapsible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-D-2940	- Dampeners, Fluid Pressure, Gage Protection.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.

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MIL-G-5572	- Gasoline, Aviation: Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-H-8794	- Hose, Rubber, Hydraulic, Fuel, and Oil Resistant.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/9	- Aircraft Loading Dataplate.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-C-23044	- Chest, Collapsible Tank.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-45150	- Chassis, Trailer, 2-Wheel, Cart Type, 1/4 to 3 1/2 Ton.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited; Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-F-52308	- Filter Element, Fluid Pressure.
MIL-F-52666	- Filter-Separator, Liquid Fuel: Frame Mounted, 350 GPM Capacity.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-V-58039	- Valves, Gate, Rising Stem, Double Acting, Aluminum.
MIL-M-82002	- Manifold and Accessory Assemblies, Portable Fuel Distribution Systems, Quick-Disconnect Hose Coupling Type.
MIL-T-82123	- Tank, Fabric, Collapsible: 20,000 Gallon, Petroleum.
MIL-H-82127	- Hose Assembly, Rubber (Synthetic): Fuel, Discharge, Collapsible.
MIL-F-82165	- Fuel Monitor Assembly: 600 Gallons Per Minute, Skid-Mounted.
MIL-M-82180	- Meter Assemblies, Skid-Mounted: Petroleum Fuels.
MIL-S-82183	- Strainer Assembly, Petroleum Fuels: 6-Inch, Frame-Mounted, 600 Gallons Per Minute.

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MIL-R-82184	- Regulator Assembly, Skid-Mounted: 6-Inch.
MIL-F-82191	- Fuel Hose Evacuation Kits.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS27023	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Flanged, Type IV.
MS27027	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Flanged, Type VIII.
MS27028	- Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029	- Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.
MS27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (in./mm).

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MS35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series, (in./mm).
MS35643	- Gasket: Synthetic Rubber, Cap, Fuel Tank.
MS35644	- Filler Neck: Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35829	- Hinge, Butt, Continuous (Piano) .075 Thickness, .187 & .250 Pin Dia.
MS51967	- Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated, UNC-2B (in./mm).
MS53028	- Chassis, Trailer: 3/4 Ton, 2 Wheel, Military Design, M116A1.
MS87006	- Hook, Chain-S.
MS90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.

DRAWINGS

U.S. MARINE CORPS

DL77C5208A0000	- Amphibious Assault Fuel System.
DL77C5209A0000	- Tactical Airfield Fuel Dispensing System.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B40.1 - Gauges - Pressure and Vacuum, Indicating Dial Type-Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM B 173 - Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 2130 Keith Building, Cleveland, OH 44115.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample system(s), assembly(ies), or component(s) for first article inspection and approval (see 4.4 and 6.3).

3.2 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3 Design and construction. The fuel receiving, storing, and dispensing systems shall be in accordance with the Data Lists DL77C5208A0000 for type I and DL77C5209A0000 for type II (see 1.2, 3.3, and 6.9) and as

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specified herein. These drawings are end product drawings and no deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the system(s), assembly(ies), or component(s). Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative. The following general requirements shall apply:

a. Type I Amphibious Assault Fuel System (AAFS). The type I AAFS is an assemblage of components and accessories having a nominal storage capacity of 600,000 gallons of fuel in collapsible fabric petroleum tanks. The system is configured normally for an ocean beach unloading area (see 1.2, 6.9, and Figure 1).

b. Type II Tactical Airfield Fuel Dispensing System (TAFDS). The type II TAFDS is an assemblage of components and accessories having a nominal storage capacity of 120,000 gallons of fuel in collapsible fabric petroleum tanks. The system is configured for installation at a tactical airfield (see 1.2, 6.9, and Figure 1).

3.3.1 Assemblies.

3.3.1.1 Fuel adapting assembly (FAA). Unless otherwise specified (see 6.2.1), the FAA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0027) and as specified herein.

3.3.1.2 Drum unloading assembly (DUA). Unless otherwise specified (see 6.2.1), the DUA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0030) and as specified herein.

3.3.1.3 Beach unloading assembly (BUA). Unless otherwise specified (see 6.2.1), the BUA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0040) and as specified herein.

3.3.1.4 Fuel dispensing assembly (FDA). Unless otherwise specified (see 6.2.1), the FDA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0050) and as specified herein.

3.3.1.5 Boost pump assembly (BPA). Unless otherwise specified (see 6.2.1), the BPA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0060) and as specified herein.

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3.3.1.6 Tank farm assembly (TFA). Unless otherwise specified (see 6.2.1), the TFA shall be furnished in accordance with the applicable drawings (ref. 77C5208A0070) and as specified herein.

3.3.2 Components.

3.3.2.1 Fuel storage tank, chest, and emergency repair kit. Unless otherwise specified (see 6.2.1), the fuel storage tank, chest, and emergency repair kit in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-T-82123 and referenced applicable drawings.

3.3.2.2 Filter-separator. Unless otherwise specified (see 6.2.1), the 350-gallons-per-minute (gpm) filter-separator in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-F-52666 and referenced applicable drawings (ref. 97403-TA13217E9320).

3.3.2.3 Filter elements. Unless otherwise specified (see 6.2.1), the filter elements in the quantities specified (see 6.2.1) shall be furnished in accordance with MIL-F-52308.

3.3.2.4 Fuel monitor assembly. Unless otherwise specified (see 6.2.1), the fuel monitor assembly in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-F-82165 and referenced applicable drawings.

3.3.2.5 Fuel meter assembly. Unless otherwise specified (see 6.2.1), a size 2- or 3-inch fuel meter assembly in the size and quantity specified (see 6.2.1) shall be furnished in accordance with MIL-M-82180 and referenced applicable drawings.

3.3.2.6 Fuel strainer assembly. Unless otherwise specified (see 6.2.1), the fuel strainer assembly in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-S-82183.

3.3.2.7 Pressure regulator assembly. Unless otherwise specified (see 6.2.1), the pressure regulator assembly in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-R-82184 and referenced applicable drawings.

3.3.2.8 Hose assemblies.

3.3.2.8.1 Suction hose assembly. Unless otherwise specified (see 6.2.1), the fuel suction hose assembly in the length, size, and quantity specified (see 6.2.1) shall be furnished in accordance with MIL-H-370, type II, class 1, style A, except that static wires shall not be included.

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3.3.2.8.2 Discharge hose assembly. Unless otherwise specified (see 6.2.1), the fuel discharge hose assembly in the length, size, and quantity as specified (see 6.2.1) shall be furnished in accordance with MIL-H-82127.

3.3.2.9 Pipe culvert. Unless otherwise specified (see 6.2.1), the pipe culvert shall be furnished in sections specified (see 6.2.1) comprised of three units of 40 linear feet in accordance with MIL-P-236, type I, class 1, size 10-inch. The bolts and nuts shall be of electro-galvanized material.

3.3.2.10 Fuel nozzles.

3.3.2.10.1 Fuel servicing nozzle. Unless otherwise specified (see 6.2.1), the fuel servicing (gravity) nozzle in the quantity specified (see 6.2.1) shall be furnished in accordance with the applicable drawing (ref. 77C5208A0048).

3.3.2.10.2 Fuel pressure nozzles. Unless otherwise specified (see 6.2.1), a size 2- or 4-inch pressure fuel servicing locking-type nozzle in the size and quantity specified (see 6.2.1) shall be furnished in accordance with the applicable drawing (ref. 77C5208A0015).

3.3.2.11 Dry-break check valve. Unless otherwise specified (see 6.2.1), the 3-inch dry-break check valve in the quantity specified (see 6.2.1) shall be furnished in accordance with the applicable drawing (ref. 77C5209A0002).

3.3.2.12 Tank chest. Unless otherwise specified (see 6.2.1), the tank chest in the quantity specified (see 6.2.1) shall be furnished in accordance with MIL-C-23044 and referenced applicable drawings.

3.3.2.13 General repair and fuel hose evacuation kits.

3.3.2.13.1 General repair kit. When specified (see 6.2.1), a complete general repair kit, or items as required and in the quantity specified (see 6.2.1), shall be furnished.

3.3.2.13.2 Fuel hose evacuation kits. Unless otherwise specified (see 6.2.1), a size 4- or 6-inch fuel hose evacuation kit in the size and quantity specified (see 6.2.1) shall be furnished in accordance with MIL-F-82191 and referenced applicable drawings (ref. 77C5208A0032 for 4-inch size and 77C5208A0046 for 6-inch size).

3.3.2.14 Manifold assembly. Unless otherwise specified (see 6.2.1), a size 4- or 6-inch manifold assembly shall conform to Figure 2 of MIL-M-82002, type II, and as specified herein. The size 4- or 6-inch

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manifold assembly shall have five nominal 4-inch or five nominal 6-inch flanged ports. The flange thickness, boltholes, bolt circle, and tolerance shall be as specified in MIL-M-82002 and compatible with the valves furnished in accordance with MIL-V-58039, type I, size as applicable. The manifold shall be assembled with valves, couplings, caps, and plugs as specified in Table I. The manifold and frame assembly shall be fabricated of aluminum as specified in Figure 2 of MIL-M-82002, and shall:

a. Provide rollover protection in all directions to all valves (when in the open position) and components (when tested) as specified in MIL-M-82002.
b. Provide slinging and tiedown provisions for a gross weight of not less than 200 pounds for size 4-inch and not less than 400 pounds for size 6-inch manifold assemblies. The slinging and tiedown provision shall conform to MIL-STD-209, class 3, for type III equipment. The attachments shall be capable of being used for towing requirements.
c. Have storage provisions for the ground rod and ground wire assemblies to prevent loss or damage during lifting, towing, and rollover tests in accordance with MIL-M-82002 and as specified in 4.6.2.1.
d. Have towing capability with not less than a 3-inch ground clearance without sustaining cracks, stress damage, or permanent deformation when tested as specified in 4.6.2.1.1.
e. Be capable of being lifted by the slinging and tiedown provisions and quick-dropped without sustaining cracks, stress damage, or permanent deformation when tested as specified in 4.6.2.1.2.
f. Be capable of being dropped on a sand-covered concrete surface without sustaining cracks, stress damage, irregularities, or permanent deformation when tested as specified in 4.6.2.1.3.

TABLE I. Size 4- and 6-inch manifold assembly.

Size 4-inch manifold assembly									
Valves		Coupling halves				Coupling cap and plug[1/]			
Size	Qty.	Male	Qty.	Female	Qty.	Cap	Qty.	Plug	Qty.
4-inch	5	MS27023-17	3	MS27027-17	2	MS27028-17	3	MS27029-17	2
Size 6-inch manifold assembly									
Size	Qty.	Male	Qty.	Female	Qty.	Cap	Qty.	Plug	Qty.
6-inch	5	MS27023-19	3	MS27027-19	2	MS27028-19	3	MS27029-19	2

[1/] The sash chain furnished with each cap and plug shall have not less than 19 links and a hook conforming to MS87006-22 for attachment to the respective coupling.

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3.3.2.14.1 Performance characteristic. The hydrostatic pressure test shall be conducted without evidence of leaks, irregularities, or damage when tested as specified in 4.6.2.1.4.

3.3.2.14.2 Fasteners. Fasteners such as capscrews, nuts, and flat and lock-type washers of the sizes required for each manifold assembly shall conform to MS90725, MS27183, MS35338, and MS51967, respectively. The fasteners shall be cadmium plated.

3.3.2.14.3 Ground rod assembly. Each manifold assembly shall be furnished with a ground rod conforming to W-R-550, type III, class B, 5/8 inch in diameter, except that three 2-foot plus or minus 1/16-inch long sections, three couplings, and one driving stud shall be furnished in lieu of the specified 3-foot sections, clamps, ground cable, and ground terminal.

3.3.2.14.4 Ground wire assembly. Each manifold assembly shall be provided with a ground wire assembly conforming to Figure 2.

3.3.2.15 Pump and diesel engine assembly. Unless otherwise specified (see 6.2.1), the pump and diesel engine assembly in the quantity specified (see 6.2.1) shall be trailer-mounted as specified on the applicable drawing (ref. 77C5208A0065) and as specified herein.

3.3.2.15.1 Pump assembly. The pump assembly shall have one each 6-inch suction and discharge port plus two each 4-inch suction and discharge ports. The pump shall be a single stage, centrifugal-type which is self-priming after the pump case has initially been filled. The pump shall prime within 2 minutes after initial filling and pump all types and grades of fuels conforming to MIL-G-3056, MIL-G-5572, MIL-T-5624, and VV-F-800 having a specific gravity from 0.67 to 0.85 as specified in 4.6.2.2.3.2, 4.6.2.2.3.3, and 4.6.2.2.3.4.

3.3.2.15.2 Impeller balancing. The impeller shall be dynamically balanced. The out-of-balance condition shall be not more than 0.25 ounce-force inch at the operating speed of the pump as specified in 4.6.2.2.2.

3.3.2.15.3 Hydrostatic pressure. The pump shall not leak or show permanent deformation when subjected to an internal hydrostatic pressure equal to 1-1/2 times the shutoff head developed at full throttle operation when tested as specified in 4.6.2.2.3.1.

3.3.2.15.4 Performance characteristics. The pump assembly, when operating at equivalent standard sea level atmospheric conditions and an equivalent 10-foot static suction lift, shall perform as follows:

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a. Deliver not less than 600 gpm of hydrocarbon fuel with a specific gravity of 0.75 against a total head (H) of 340 feet, at a maximum net positive suction head (NPSH) of 9 feet at 2,400 plus or minus 50 revolutions per minute (r/min) as specified in 4.6.2.2.3.3.
b. Deliver not less than 900 gpm of hydrocarbon fuel with a specific gravity of 0.75 against a total H of 209 feet, at a maximum NPSH of 12 feet at 2,000 plus or minus 50 r/min as specified in 4.6.2.2.3.3.
c. Operate, after initial priming, for not less than 100 hours when tested as specified in 4.6.2.2.4.
d. The pumping unit shall perform as specified herein at any ambient temperature from minus 25° F to plus 125° F.
e. The pumping unit shall not be damaged by storage under the following conditions:

(1) High temperature -- plus 155° F for not less than 4 hours.
(2) Low temperature -- to minus 65° F for not less than 12 hours without benefit of solar radiation.

The pump assembly shall exhibit no instability in pressure, flow rate, or any engine malfunction when tested as specified in 4.6.2.2.4.

3.3.2.15.5 Performance curve plate. A performance curve plate, 8 x 10-1/2 plus or minus 1/16 inches, shall be permanently attached to the pump assembly by screws, rivets, or bolts in a location where the plate will be visible and legible. The chart shall depict both performance curves and rating points for the two speeds given in the performance requirements specified in 3.3.2.15.4. The data shall be derived from the results of the first article tests specified in 4.4.3. The plate material shall be as specified in 3.7.2.

3.3.2.15.6 Engine. The pump assembly shall be driven by a diesel engine as specified herein and on the applicable drawing. The engine shall be a liquid-cooled, two-stroke cycle, multiple cylinder-type with a continuous brake horsepower output rating to equal or exceed the horsepower demands of the pump at performance rating specified in 3.3.2.15.4 a and b. The engine shall be equipped with, but not be limited to, the following:

a. Electrical engine controls to include a master ignition switch to energize the starter, oil pressure gage, water temperature gage, battery/generator gage, panel lights, panel light switch, solenoid valve and safety control devices.
b. Manual engine controls to include a waterproof pushbutton starter switch, micrometer throttle control and cable (to select governed speed from idle to maximum governed speed),

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shutdown control (pull-to-stop), safety override control (control overrides oil pressure safety feature during starting phase of operation), and cold weather starting aid control.

c. A vacuum-type cooling fan.

d. A radiator-type cooling system.

e. Fuel tank of not less than 20-gallon capacity, equipped with a filler neck conforming to MS35644-1, a tank cap conforming to MS35645-1, and a gasket conforming to MS35643-1. The tank shall be equipped with a mechanically actuated fuel gage graduated from empty to full.

f. Two three-way fuel selector valves to permit engine operation from the fuel tank or an auxiliary fuel service.

g. Two corrosion-resisting circular face plates mounted on the fuel selector valves with markings of "OFF", "UNIT TANK", and "AUXILIARY".

h. Two 10-foot-long fuel hoses conforming to MIL-H-8794 sized to permit unrestricted fuel flow, one on each auxiliary leg of the selector valves and with fittings compatible with the three-way selector valves.

i. Mechanical automatic low oil pressure, high water temperature and overspeed safety controls. The system must incorporate a means to bypass the low oil pressure shutdown during starting operation.

j. The instrument panel shall be shock mounted.

k. Two shock- and vibration-resistant pressure gages of non-oxidizing material conforming to ANSI B40.1. The discharge gage shall have a range from zero to 200 pounds-force/inch^2 gage (lb_f/in^2) pressure. The compound pressure/suction gage shall have a range from zero to 100-lb_f/in^2 gage pressure and a range from zero to 30 inches of mercury, vacuum. Each gage shall be accurate to within 2 percent of the total gage reading. The dials shall be white with black calibration lines and figures, and shall be not less than 4-1/2 inches in diameter. The gages shall be located on the instrument panel and shall be provided with a fluid-pressure dampener conforming to MIL-D-2940, type II, class 1, composition optional. Shutoff cocks shall be provided in gage lines. All gage lines shall be located and protected to prevent damage by personnel while handling during operation and service, and shall have provisions for draining to prevent freezing.

3.3.2.15.7 Coupling. The pump shall be connected to the engine with a flywheel-type, dry-type, flexible coupling. The coupling shall transmit full engine torque and shall be replaceable without disconnecting the pump or engine from the frame or disconnecting the inlet or outlet piping.

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3.3.2.15.8 Battery box. A battery box shall be fabricated in accordance with MIL-STD-1410.

3.3.2.15.9 Storage box. The pump and engine assembly shall be equipped with a weatherproof storage box made of merchant-type steel not thinner than 15-gage thickness. The storage box shall be provided with a lid, of the overlapping-type, to prevent entry of rain or dirt. The lid shall be secured to one edge of a lid by a continuous hinge conforming to MS35829, size as applicable, and shall be held in the closed position by not less than two fasteners. The storage box shall be of a size capable of storing battery electrolytes, trailer pintle hook and reflectors (when removed for shipment), minor operating and maintenance tools, and operating and maintenance manuals.

3.3.2.15.10 Ground rod and ground wire assembly. Each pump/engine assembly shall be furnished with a ground rod and ground wire assembly as specified in 3.3.2.14.3 and 3.3.2.14.4, respectively.

3.3.2.15.11 Slinging and tiedown provisions. Permanently affixed to the trailer chassis shall be multipurpose slinging and tiedown provisions which shall be compatible with the transport and lifting mode requirements, and MIL-STD-209, class 3, for type III equipment when tested as specified in 4.6.2.2.6.

3.3.2.15.12 Trailer chassis assembly. Unless otherwise specified (see 6.2.1), the pump and engine assembly shall be mounted on a Government-furnished trailer chassis as specified in 3.9. The contractor may be authorized by the Government to modify the trailer chassis to accommodate the engine and pump assembly, in which case the modifications shall not be detrimental to the vehicle characteristics specified in MS53028 and MIL-C-45150. Upon completion of all modifications, and with the pump and engine assembled onto the trailer chassis, the trailer assembly shall be towed by a Government-loaned vehicle (see 3.10) and tested as specified in 4.6.2.2.7.

3.3.2.15.13 Trailer rear support legs. Two rear support legs shall be provided to prevent the pump unit from tipping over when hose or pipe is coupled to the pump or when the pump or bearing housing is being filled or serviced by personnel. The legs shall be capable of being securely locked in both the retracted and extended positions. The locking devices and the legs shall be captive-chained to the pump unit frame to prevent loss. The legs shall be capable of being adjusted by hand in 2-inch increments from the fully retracted position to the fully extended positions, and vice versa. When in the fully retracted position, the support legs shall not protrude more than 2 inches below the bottom of the side members of the frame. The ground-bearing surfaces of the rear support legs shall contain not less than 16 square inches per leg.

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3.3.3 Human factors engineering. The pump and engine assembly shall conform to the controls, maintainability, and hazards and safety human factors engineering criteria as specified in MIL-STD-1472 (see 4.6.2.2.8).

3.3.4 Safety. All belts, gears, pulleys, shafts, the flywheel, and other moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel, or which create a fire hazard, shall be fully guarded or insulated. The fuel tank shall be located in a manner which will not allow spills or overflows to run on the engine, exhaust pipes, or electrical equipment. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Live parts of wiring and electrical equipment shall be guarded and grounded to protect all persons and objects from electrical shock hazard. Exhaust or discharge from equipment shall be directed so that they do not endanger personnel (see 4.6.2.2.8).

3.3.5 Lubricants. The pump/engine assembly shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Lubricating oil:	MIL-L-2104 or MIL-L-46167.
Grease:	MIL-G-10924 or MIL-G-23827.

3.3.6 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.3.7 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.3.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see c, below), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant, or which are hermetically sealed, need not be treated.

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b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.
c. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with rule 66, Air Pollution Control District, County of Los Angeles, CA.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.3.9 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.3.10 Castings and forgings. All parts, components, and assemblies of the pump which include castings and forgings shall be clean of harmful extraneous material such as gas holes, scale, flux, and investment material. Rework shall be limited to procedures which do not reduce mechanical properties, or affect function.

3.3.11 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.3.12 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not less

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than two full threads nor more than the next incremental length. Prior to assembling, threads on all screwed fittings shall be coated with sealing compound conforming to MIL-S-7916. Unless otherwise specified herein and on applicable drawings, all screw threads shall be in accordance with NBS Handbook H28.

3.3.13 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.3.14 Connections, hoses, piping, and cables. All connections, hoses, piping, and cables shall be supported to prevent damage caused by vibration, chafing, or sharp bends.

3.3.15 Machine work. Tolerances and gages for metal-fits shall conform to the limits herein, on applicable drawings, and to the standards of the industry.

3.4 Maintainability.

3.4.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.2.9.

3.4.2 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.5 Electromagnetic interference. The electromagnetic interference emission characteristics of the engine assembly shall conform to MIL-STD-461, class IIIC when tested as specified in 4.6.2.2.10.

3.6 Treatment and painting. Unless otherwise specified in referenced specifications or drawings, all exterior surfaces normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the finish coat shall be in accordance with MIL-E-52798 (forest green).

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3.7 Identification, instruction, and shipping plates.

3.7.1 Identification marking. Each pump assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type I, style 3, composition C, of type optional, grade B, class 1 material. The plates shall be attached to the pump assembly by screws, rivets, or bolts in a place where the plate will be visible and legible but not interfere with the operation and maintenance of the pump assembly.

3.7.2 Instruction plate. Unless otherwise specified (see 6.2.1) each pump assembly shall be equipped with an instruction plate conforming to MIL-P-514, type III, composition C, of type optional, grade B, class 1 material. The plate shall include all warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the pump assembly. The plate shall include the warning, "NO SMOKING WITHIN 50 FEET." The plate shall be attached to the pump assembly by screws, rivets, or bolts in a place where the plate will be visible and legible, but not interfere with the operation and maintenance of the pump assembly.

3.7.3 Shipping data plate. Each pump and engine assembly shall be provided with an aircraft loading data plate conforming to Figure 3 of MIL-P-15024/9, type H, style III. The plate shall be attached to the pump and engine assembly as specified in MIL-P-15024 by screws, rivets, or bolts where the plate will be visible and legible.

3.7.4 Identification marking of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump or by plates specified in 3.7.2 permanently secured in place by means of metal fasteners.

3.8 Functional evaluation. When specified (see 6.2.1) a complete type I or type II fuel system shall be assembled and evaluated for any leaks, irregularities, or permanent deformation during functional testing as specified in 4.6.2.2.11.

3.9 Government-furnished property. Unless otherwise specified (see 6.2.1), the pump and engine assembly shall be mounted on a Government-furnished chassis, trailer, 3/4-ton, 2-wheel, M116A1 in accordance with MS53028-1 (see 6.6).

3.10 Government-loaned property. Unless otherwise specified (see 6.2.1), the Government will loan a truck, cargo, 2-1/2 ton, M35A2 for mobility testing (see 6.7).

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3.11 Workmanship. The workmanship shall be of a quality to assure that the fuel system(s) and assembly(ies), and component(s) thereof, are free from any defects that compromise, limit, or reduce the fuel system(s), assembly(ies), or components' capabilities as specified herein. Workmanship shall be such as to provide maximum protection to the assembly(ies) and component(s) by application of the requirements specified herein and referenced documents, to prevent corrosion, deterioration, and mechanical or physical damage during handling, shipment, and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Component and material inspection (see 4.3).
b. Disassembly inspection (see 4.3.1).
c. First article inspection (see 4.4).
d. Quality conformance inspection (see 4.5).
e. Packaging inspection (see 4.7).

4.3 Component and material inspection. Components and materials used in the manufacturing, testing, and packaging of the fuel system, assemblies, or components shall be inspected by the contractor in accordance with the applicable referenced documents. Nonconformance to the applicable referenced documents shall be cause for rejection.

4.3.1 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced documents in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.4 First article inspection.

4.4.1 Sampling. From each first article fuel system, a random sample of one or more of each component furnished by the contractor shall be selected for examination and tests.

4.4.2 Examination. The first article samples selected in accordance with 4.4.1 shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. The first article samples selected in accordance with 4.4.1 shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspections.

4.5.1 Lot. For the purpose of sampling, a lot shall consist of one fuel system (either type I or type II), associated fuel assemblies, or components submitted for inspection at the same time and place.

4.5.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.3 Examination.

4.5.3.1 Individual. Each component shall be examined for critical and major defects specified in 4.6.1. Presence of any critical or major defect shall be cause for rejection of the system(s), assembly(ies), or component(s) offered.

4.5.3.2 Samples. Samples selected in accordance with 4.5.2 shall be examined for the minor defects specified in 4.6.1. Acceptable quality level (AQL) shall be 4.0 percent defective.

4.5.4 Tests. Samples selected in accordance with 4.5.2 shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.6 Inspection procedures.

4.6.1 Examination. The fuel system(s), assembly(ies), or component(s), including the manifold assembly and the pump and engine assembly, shall be examined as specified herein for the following defects:

Critical

1. Porosity, gas holes, nonmetallic inclusions, internal shrinkage, cracks, and tears in castings.

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Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Any fuel system not complete or not as specified, when applicable.
104. Any assembly not complete or not as specified, when applicable.
105. Fuel storage tanks, chests, and emergency repair kit not as specified.
106. Filter elements not as specified, when applicable.
107. Fuel monitor assembly not as specified, when applicable.
108. Fuel meter assembly not as specified, when applicable.
109. Fuel strainer assembly not as specified, when applicable.
110. Pressure regulator assembly not as specified, when applicable.
111. Suction or discharge hose assemblies not as specified, when applicable.
112. Pipe culvert not as specified, when applicable.
113. Fuel nozzles not as specified, when applicable.
114. Dry-break check valve not as specified, when applicable.
115. Tank chest not as specified, when applicable.
116. General repair kit not as specified, when applicable.
117. Fuel hose evacuation kit not as specified, when applicable.
118. Manifold assembly not as specified, when applicable.
119. Ground rod assembly not as specified.
120. Ground wire assembly not as specified.
121. Pump and engine assembly not as specified, when applicable.
122. Coupling not as specified, when applicable.
123. Battery box not as specified, when applicable.
124. Storage box not as specified, when applicable.
125. Slinging and tiedown provisions not as specified, when applicable.
126. Trailer chassis not as specified, when applicable.
127. Trailer rear support legs not as specified, when applicable.
128. Safety devices, fuel tank, electrical wiring, hot exhaust discharge not as specified, when applicable.
129. Covers and plates not furnished as specified, when applicable.
130. Lubricants not as specified.
131. Lubrication fittings not as specified.
132. Electrical circuitry not protected from fungus growth as specified, when applicable.
133. Treatment and painting not as specified, when applicable.
134. Workmanship not as specified.

Minor

201. Fasteners not as specified or not cadmium plated.
202. Maintenance support elements not as specified, when applicable.

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203. Identification marking not as specified.
204. Instruction plate not as specified, when applicable.
205. Shipping data plate not as specified.

4.6.2 Tests.

4.6.2.1 Manifold assembly.

4.6.2.1.1 Towing. The manifold assembly, complete with all components and parts, and filled with a test media shall be towed for not less than 500 feet across a hard, rough surface at not less than 5 miles per hour (mi/hr). Upon completion of this test, the manifold assembly shall be examined for cracks, stress damage, or permanent deformation. Nonconformance to 3.3.2.14 shall constitute failure of this test.

4.6.2.1.2 Lifting. The manifold assembly complete with all components and parts shall be lifted by the lifting and tiedown attachments to a height not less than 6 feet and held suspended for not less than 15 minutes. At the end of 15 minutes, the assembly shall be dropped not less than 4, nor more than 5, feet and quick-stopped by the lifting cable. Upon completion of this test, the manifold assembly shall be examined for cracks, stress damage, or permanent deformation. Nonconformance to 3.3.2.14 shall constitute failure of this test.

4.6.2.1.3 Drop. The manifold assembly complete with all components and parts, and filled with a test media, shall be lifted not less than 24 inches and free dropped onto a surface consisting of 3 inches of packed sand over a concrete floor. The manifold assembly shall be examined for cracks, breaks, irregularities, or permanent deformation which would render the manifold assembly unsuitable for its intended use. After the examination, the manifold assembly shall again be subjected to the hydrostatic pressure test. Nonconformance to 3.3.2.14 shall constitute failure of this test.

4.6.2.1.4 Performance characteristic. The manifold assembly shall be subjected to the hydrostatic pressure test of MIL-M-82002 for not less than 10 minutes with plugs and caps removed and valves closed. Nonconformance to 3.3.2.14.1 shall constitute failure of this test.

4.6.2.2 Pump and engine assembly.

4.6.2.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury and liquid fuel at 60° F having a specific gravity of 0.75 and vapor pressure of 4.0 lb_f/in^2. Water at a temperature between 60° F and 80° F may be used as the test fluid.

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4.6.2.2.2 Preassembly test. During assembly of the pump unit, determine the out-of-balance characteristics of the impeller. Out-of-balance of more than 0.25 ounce-force-inch or nonconformance to 3.3.2.15.2 shall constitute failure of this test.

4.6.2.2.3 Assembled tests.

4.6.2.2.3.1 Hydrostatic. Subject the pump to a hydrostatic pressure equal to 1-1/2 times the shutoff head obtained at full throttle operation. Leakage or nonconformance to 3.3.2.15.3 shall constitute failure of this test.

4.6.2.2.3.2 Priming. The pump assembly shall be primed and operated under conditions specified herein. Nonconformance to 3.3.2.15.1 shall constitute failure of this test.

4.6.2.2.3.3 Capacity. The pump assembly shall be primed and operated to determine the deliverable volume, head capacities, NPSH, engine rpm, and efficiency. Nonconformance to 3.3.2.15.4 shall constitute failure of this test.

4.6.2.2.3.4 Check valve. After the pump has been completely primed, subject the check valve to a maximum test pressure equal to the maximum total head. Shut off the pump, drain the suction piping and allow the pump to set for not less than 30 minutes. Restart the pump, and determine that the pump has primed itself. Self-priming of the pump is an indication that the valve prevents siphoning of fuel from the pump case. Inability of the pump to prime itself, leakage, or permanent deformation of any component of the valve shall constitute failure of this test.

4.6.2.2.4 Operational (100-hour). Prime the pump in accordance with 4.6.2.2.3.2 using a static suction lift equivalent to 10 feet. Operate the pump at the head-capacity condition requiring the maximum brake horsepower determined during test under 4.6.2.2.3.3 continuously for 100 hours. During operation, only minor adjustment to the engine and discharge valving may be performed. A maximum shutdown time of 15 minutes is allowable for normal servicing and maintenance and examination during each 12-hour period. During operation, determinations shall be made as to whether there is leakage through the pump case, pump flange, piping, fittings, gages, or at the pump shaft seal, and whether there is a malfunction of any component. Midway through the test, determine whether all the safety devices are functioning. Evidence of leakage, malfunction of any component, inability of the pump to prime within 2 minutes or to deliver the capacity established at the beginning of the test, inability of the safety devices to perform the function intended, or inability of the pump unit to operate continuously (except for authorized shutdown) shall constitute failure of this test. Nonconformance to 3.3.2.15.4 shall constitute failure of this test.

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4.6.2.2.5 Production.

4.6.2.2.5.1 Pumping assembly. Test the pump unit as specified in 4.6.2.2.4 except that the operating time shall be 30 minutes and shall be at the flow rate of 600 gpm. Nonconformance to 3.3.2.15.4 shall constitute failure of this test.

4.6.2.2.5.2 Production engine. Prior to assembly, each engine shall be tested in accordance with MIL-STD-1400, test method series 4000, or the engine manufacturer's standard production run-in test, provided it equals or exceeds MIL-STD-1400, test method series 4000. Failure to produce the power specified in MIL-STD-1400, test method series 4000, shall constitute failure of this test.

4.6.2.2.5.3 Production control. One engine out of each 50, or fraction thereof, shall be tested in accordance with MIL-STD-1400, test method series 3000. Failure of the engine to produce horsepower in accordance with the engine manufacturer's published maximum brake horsepower curve shall constitute failure of this test.

4.6.2.2.6 Lifting and tiedown attachments. Lifting and tiedown attachments shall be subjected to static pull tests equivalent to those specified in MIL-STD-209, class 3. Nonconformance to 3.3.2.15.11 shall constitute failure of this test.

4.6.2.2.7 Trailer chassis. The trailer chassis, if modified, shall be examined and tested with complete payload as specified for initial production test requirements in accordance with MIL-C-45150, Table V. Nonconformance to 3.3.2.15.12 shall constitute failure of this test.

4.6.2.2.8 Human factors engineering. The pump and engine assembly controls, maintainability, and hazards and safety factors shall be evaluated for conformity. Nonconformance to 3.3.3 and 3.3.4 shall constitute failure of this test.

4.6.2.2.9 Maintenance support evaluation. Maintenance operations required during testing shall be accomplished to determine conformance to 3.4.1. Failure of the support elements to maintain the pump and engine assembly, as specified, shall constitute failure of this test.

4.6.2.2.10 Electromagnetic interference. The first-produced pump and engine assembly shall be equipped for the reduction of radiated electromagnetic emissions and shall be tested in accordance with MIL-STD-462, test method RE05. The contractor shall furnish the contracting officer the test report required in MIL-STD-461. Disapproval of the reports and nonconformance to 3.5 shall constitute failure of this test.

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4.6.2.2.11 Functional. The samples selected in accordance with 4.5.2 shall be combined to form a continuous fuel system or assembly. Assembling and disassembling of the test samples shall be accomplished by hand without the use of special tools. The assembly of the test sample shall be subjected to a fuel pressure of not less than 150 lb_f/in^2. While under pressure, the entire system, assembly(ies), and components shall be examined for leaks, irregularities, or permanent deformations in any hose, valve, or fitting, and for slippage or pullout of the hose or tank fitting. Each valve and nozzle shall be individually activated. Inability to assemble the fuel system or assembly by hand, evidence of leakage, irregularities, or permanent deformation of any sample, or inability to operate any functional item shall constitute failure of this test. The volume of test fuel or water shall be such as to supply a flow rate of not less than 600 gpm at not less than 150 lb_f/in^2. The pressure may be developed by means of throttling gate valves located at strategic locations. Nonconformance to 3.8 shall constitute failure of this test.

4.7 Packaging inspection.

4.7.1 First article pack inspection.

4.7.1.1 Examination. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The first article pack of crated components, packed for level A, shall be subjected to the railroad-car test specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and brakes set, at a speed of not less than 10 miles, nor more than 11 mi/hr. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the first article pack and shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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135. Fuel storage tanks not preserved and packaged in accordance with the referenced document as specified for level A.
136. Filter-separators not preserved and packaged in accordance with the referenced document as specified for level A.
137. Filter elements not preserved and packaged in accordance with the referenced document as specified for level A.
138. Extra O-rings and extra sight gage gaskets not preserved and packaged in accordance with the referenced document as specified for level A.
139. Fuel monitor assemblies not preserved and packaged in accordance with the referenced document as specified for level A.
140. Fuel meter assemblies not preserved and packaged in accordance with the referenced document as specified for level A.
141. Fuel strainer assemblies not preserved and packaged in accordance with the referenced documents as specified for level A.
142. Fuel pressure regulator assemblies not preserved and packaged in accordance with the referenced document as specified for level A.
143. Hose assemblies not preserved and packaged in accordance with the referenced document as specified for level A.
144. Pipe culvert not preserved in accordance with the referenced document as specified for level A.
145. Fuel servicing and pressure nozzles not preserved and packaged in accordance with the referenced document as specified for level A.
146. Dry-break check valves not preserved and packaged in accordance with the referenced document as specified for level A.
147. General repair kits not preserved and packaged in accordance with referenced document and specified method as specified for level A.
148. Fuel hose evacuation kits not preserved and packaged in accordance with the referenced document as specified for level A.
149. Manifold assemblies not preserved and packaged in accordance with the referenced document as specified for level A.
150. Pump and diesel engine assembly not disassembled, preserved and packaged as specified for level A.
151. Gaskets and sealing compound not preserved and packaged in accordance with the referenced document and specified method as specified for level A.
152. Components not specifically specified not preserved and packaged as specified for level A.
153. Consolidation not as specified for level A.
154. Components not preserved and packaged in accordance with the referenced document for commercial.
155. Engine crankcase and lubricant reservoirs not filled as specified for commercial.

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156. Cooling system not filled as specified for commercial.
157. Pumping assembly not protected with a housing as specified and not secured as specified for level A.
158. Storage tanks and tank chests not packed as specified for level A.
159. Filter-separators not packed in accordance with the referenced document as specified for level A.
160. Packing of components in crates not as specified for level A.
161. Packing not as specified for commercial.
162. Marking missing, illegible, incorrect, or incomplete for level A or commercial.

4.7.2.4 Tests. The contractor shall furnish a first article pack for tests within the time frame specified (see 6.2.1), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirement of this specification. Tests shall be as specified herein and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article fuel system, component, or assembly, or a production fuel system, component, or assembly. If the first article fuel system, component, or assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article fuel system, component, or assembly and the production fuel system, component, or assembly.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Fuel storage tank. Each fuel storage tank shall be preserved and packaged in accordance with the level A requirements of MIL-T-82123.

5.1.1.2 Filter-separator. Each filter-separator shall be preserved and packaged in accordance with the requirements of MIL-F-52666.

5.1.1.3 Filter elements. Each filter element shall be preserved and packaged in accordance with the requirements of MIL-F-52666.

5.1.1.4 Extra O-rings and extra sight gage gasket. Each extra O-ring sight gage gasket shall be preserved and packaged in accordance with the requirements of MIL-F-52666.

5.1.1.5 Fuel monitor assembly. Each fuel monitor assembly shall be preserved and packaged in accordance with the level A requirements of MIL-M-82165.

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5.1.1.6 Fuel meter assembly. Each fuel meter assembly shall be preserved and packaged in accordance with the level A requirements of MIL-M-82180.

5.1.1.7 Fuel strainer assembly. Each fuel strainer assembly shall be preserved and packaged in accordance with the level A requirements of MIL-S-82183.

5.1.1.8 Fuel pressure regulator assembly. Each fuel pressure regulator assembly shall be preserved and packaged in accordance with the level A requirements of MIL-R-82184.

5.1.1.9 Hose assemblies. The hose assemblies shall be preserved and packaged in accordance with the level A requirements of MIL-P-775.

5.1.1.10 Pipe culvert. Each pipe culvert section shall be preserved in accordance with level A requirements of MIL-P-236.

5.1.1.11 Fuel servicing and pressure nozzles. Each fuel servicing and pressure nozzle shall be preserved and packaged in accordance with level A requirements of MIL-P-775.

5.1.1.12 Dry-break check valve. Each dry-break check valve shall be preserved and packaged in accordance with level A requirements of MIL-V-3.

5.1.1.13 Tank chest. The tank chests shall not require preservation and packaging.

5.1.1.14 General repair kit. The general repair kit or items as required shall be preserved and packaged in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.15 Fuel hose evacuation kit. The size 4- or 6-inch fuel hose evacuation kit shall be preserved and packaged in accordance with level A requirements of MIL-F-82191.

5.1.1.16 Manifold assembly. The size 4- or 6-inch manifold assembly shall be preserved and packaged in accordance with level A requirements of MIL-M-82002.

5.1.1.17 Pump and diesel engine assembly. The pump and diesel engine assembly shall be preserved and packaged as specified in 5.1.1.17.1 through 5.1.1.17.10.

5.1.1.17.1 Disassembly. The trailer pintle hook shall be removed and placed in the storage box. The reflectors and any other components

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requiring protection against damage, pilferage, or loss, or protruding parts that increase cubage and can be removed and reinstalled without special tools or skills, shall be removed in accordance with the contractor's recommendations. Bolts, nuts, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.1.17.2 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.1.1.17.3 Pump. Interior surfaces of the pump and connecting piping shall be coated with lubricating oil conforming to MIL-L-21260, type I, grade 30. The pump shall be actuated during application of the oil to insure coating of all interior surfaces. After interior preservation is completed, the excess oil shall be drained. The draincock shall be left open. All other valves shall be closed and the dust caps and plugs installed.

5.1.1.17.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type II, method I.

5.1.1.17.5 Tires. Tires shall be inflated to 10 pounds above pressure value recommended for maximum load.

5.1.1.17.6 Repair parts. The preservative application criteria and applicable method(s) of preservation in accordance with MIL-P-116 shall be used to preserve repair parts, and the repair parts shall be placed in the storage box.

5.1.1.17.7 Operating and maintenance tools. Operating and maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40 and placed in the storage box.

5.1.1.17.8 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and placed in the storage box.

5.1.1.17.9 Storage box. The items contained in the storage box shall be cushioned in a manner to prevent damage to the items. The storage box lid shall be closed and secured.

5.1.1.17.10 Removed-components. Components which will not fit in the storage box shall be secured to the pump, in a protected location which will not increase the cubage or interfere with lifting or towing. They

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shall be secured with soft annealed wire or with strapping conforming to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2.1), strapping shall be finish B.

5.1.1.18 Caps and plugs. Aluminum caps and plugs furnished with each component shall be fitted to their respective valve or fitting.

5.1.1.19 Gaskets and sealing compound. Gaskets shall be individually preserved, in a flat position, in accordance with MIL-P-116, method IC-1 or IC-3. The gaskets and the sealing compound shall then be packaged together in a close-fitting box conforming to PPP-B-636, W6c. The sealing compound shall be secured within the box in a manner to avoid damage to the gaskets.

5.1.1.20 Other components. Components of the fuel system not specifically specified herein shall be preserved and packaged as specified for components and equipment of similar design and construction.

5.1.1.21 Consolidation. Components, except for the tanks, tank chests, pumps, and filter-separators, shall be consolidated together in boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. When practical, the boxes shall contain only components of like description. The boxes shall be of a size and configuration to most conveniently accomplish the packing specified in 5.2.1. The components shall be blocked, braced, anchored, or cushioned, as applicable, within the boxes to prevent movement or damage to the contents. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required.

5.1.2 Commercial. Components of the fuel system shall be preserved and packaged to afford protection against deterioration and damage, from the contractor to the initial destination, in accordance with MIL-STD-1188 and the following specific requirements. Fittings and hose assemblies shall be sealed as specified in 5.1.1.18. The engine crankcase and all lubricant reservoirs shall be filled to operating level with the lubricant recommended for operation. The cooling system shall be filled with a fresh, clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Pumping assembly. The trailer mounted pumping assembly shall be prepared for mobile shipment. A plywood housing consisting of two ends, two sides, and a top shall be placed over the components and extended to the trailer frame. The housing shall be constructed in accordance with the requirements of PPP-B-601, overseas type, grade B, style I,

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and shall be modified as required for the configuration to permit securing to the trailer frame. The housing shall be secured in a manner that will not interfere with lifting or towing by the use of lumber dunnage as required and not less than four steel straps. The straps shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2.1), strapping shall be finish B. No fewer than two straps shall be applied going completely around the housing from front to rear under the trailer frame and no fewer than two straps shall be applied going around the housing from side to side under the trailer frame.

5.2.1.2 Storage tanks and tank chests. Storage tanks shall be packed in accordance with the level A requirements of MIL-T-82123, utilizing the tank chests when they are procured. Tank chests procured without storage tanks shall require no packing but shall have the lids secured to prevent accidental opening.

5.2.1.3 Filter-separators. Filter separators shall be packed in accordance with the level A requirements of MIL-F-52666.

5.2.1.4 Other components. All the components of the fuel system, not specified for packing in 5.2.1.1, 5.2.1.2, and 5.2.1.3, shall be packed together in one or more crates as applicable. The crate or crates as applicable shall conform to MIL-C-3774, nailed assembly, skid-type base or MIL-C-104, type I, class 1 or 2, style a, as applicable. The contents shall be secured to the crate base in a manner to provide minimum cube. Blocking, bracing, and anchoring of the contents shall be in accordance with the appendix to the applicable crate specification and in accordance with MIL-STD-1186.

5.2.2 Commercial. The trailer-mounted pump and engine assembly shall be prepared for mobile shipment and the storage box closed and secured in a manner to prevent pilferage. All other components of the fuel system shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages, and shipping containers, shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The type I AAFS and type II TAFDS are intended to receive, store, transfer, and dispense bulk fuels such as automotive and aviation gasoline, JP-4 and JP-5 jet fuel, or diesel fuel by performing the following:

a. Receive bulk fuel from offshore tankers, railroad tank cars, or other sources for temporary storage in the collapsible fabric fuel tanks.

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b. Transfer bulk fuel from receiving units through temporary above-ground hose lines to the tank farm of collapsible fabric fuel tanks.
c. Dispense fuel from the tank farms to the receiving vehicles or aircraft through the dispensing units.
d. Process the fuel at various stages of transfer to insure delivery of acceptable fuel of the correct type, free of contamination.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type of fuel system required (see 1.2, 3.3a and b, and 6.9).
c. When first article is required for inspection and approval (see 3.1, 4.4, and 6.3).
d. How an assembly is to be furnished if other than specified (see 3.3.1).
e. When the collapsible fuel storage tanks, chests, or emergency repair kit shall not be furnished, or if furnished, quantity required (see 3.3.2.1).
f. When the filter-separator shall not be furnished, or if furnished, quantity required (see 3.3.2.2).
g. When the filter elements shall not be furnished, or if furnished, quantity required (see 3.3.2.3).
h. When the fuel monitor assembly shall not be furnished, or if furnished, quantity required (see 3.3.2.4).
i. When the fuel meter assembly shall not be furnished, or if furnished, size and quantity required (see 3.3.2.5).
j. When the fuel strainer assembly shall not be furnished, or if furnished, quantity required (see 3.3.2.6).
k. When the pressure regulator assembly shall not be furnished, or if furnished, quantity required (see 3.3.2.7).
l. When the suction hose assembly shall not be furnished, or if furnished, length, size, and quantity required (see 3.3.2.8.1).
m. When the discharge hose assembly shall not be furnished, or if furnished, length, size, and quantity required (see 3.3.2.8.2).
n. When the pipe culvert shall not be furnished, or if furnished, number of sections required (see 3.3.2.9).
o. When fuel servicing nozzle shall not be furnished, or if furnished, quantity required (see 3.3.2.10.1).
p. When fuel pressure nozzle shall not be furnished, or if furnished, size and quantity required (see 3.3.2.10.2).
q. When the dry-break check valve shall not be furnished, or if furnished, quantity required (see 3.3.2.11).

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r. When the tank chest shall not be furnished, or if furnished, quantity required (see 3.3.2.12).
s. When the general repair kit shall not be furnished, or if furnished, quantity of kits or any items of the kit required (see 3.3.2.13.1).
t. When the fuel hose evacuation kit shall not be furnished, or if furnished, size and quantity required (see 3.3.2.13.2).
u. When the manifold assembly shall not be furnished, or if furnished, size and quantity required (see 3.3.2.14).
v. When the trailer-mounted pump and engine assembly shall not be furnished, or if furnished, quantity required, or if pump and engine is to be other than trailer-mounted (see 3.3.2.15 and 3.3.2.15.12).
w. When instruction plate is not required (see 3.7.2).
x. When functional evaluation is required (see 3.8).
y. How the trailer is to be furnished if other than Government-furnished (see 3.9).
z. How the truck is to be furnished if other than Government-loaned (see 3.10).
aa. Time frame required for submission of the first article pack (see 4.7.2.4).
bb. Level of preservation and packing required (see 5.1 and 5.2).
cc. When other than finish B strapping is required (see 5.1.1.17.10 and 5.2.1.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4	First Article Inspection Procedures.	DI-T-4901
4.4	First Article Inspection Report.	DI-T-4902
4.5	Production/Acceptance Inspection Procedures.	DI-T-4903
4.6	Inspection System Program Plan.	DI-R-4803
4.6	End Item Final Inspection Record.	DI-R-4809

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisons of 7-104.55 of the Armed Services Procurement Regulation. The first article should be the first-produced item. The first article should consist of assemblies, or components required by the contract or purchase order for each type I or type II fuel system. The contracting officer should include specific instructions for all procurement instruments regarding arrangements for examination, test, and approval of the first article (see 3.1).

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the fuel system(s) in accordance with this specification (see 4.7.2.4).

6.5 Recycled material. It is encouraged that recycled or reclaimed material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.6 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.9, when applicable (see 3.9 and 6.2.1).

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10, when applicable (see 3.10 and 6.2.1).

6.8 Information figures. Figure 1, tabulating typical type I and type II fuel systems' composition, is funished for information only (see 3.3a and b).

6.9 Classification relationship. The relationship between classification of the fuel systems included in this revision of the specification, the previous edition, and superseded document is as follows (see 1.2 and 3.3a and b):

New designation	Old designation
Type I	AAFS-MIL-F-82129A(MC)
Type II	TAFDS-MIL-F-28929(MC)
	CANCELED

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6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:

Navy - MC

Project 4930-N213

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Fuel components, assemblies, and system list.

Item	Components	Spec. ref.	Assemblies[1]						Systems	
			FAA	DUA	BUA	FDA	BPA	TFA	AAFS type I	TAFDS[2] type II
1	Tank, fabric, 20,000 gallons	3.3.2.1			2		2	6	36	6
2	Filter-separator, 350 gpm	3.3.2.2				1			4	6
3	Fuel monitor assembly, 600 gpm	3.3.2.4							4	6
4	Fuel meter assy. 2-inch	3.3.2.5				6			12	12
5	Fuel meter assy. 3-inch	3.3.2.5			1	1			3	3
6	Fuel strainer assembly, 600 gpm	3.3.2.6			1				1	
7	Pressure regulator assy., 6-inch	3.3.2.7							1	
8	Hose, suction, 2-in. x 25 ft.	3.3.2.8.1		6					6	8
9	Hose, suction, 4-in. x 25 ft.	3.3.2.8.1		3	13		4	18	114	19
10	Hose, discharge, 2-in. x 25 ft.	3.3.2.8.2	2			12			28	
11	Hose, discharge, 2-in. x 50 ft.	3.3.2.8.2	2						24	36
12	Hose, discharge, 4-in. x 50 ft.	3.3.2.8.2	6	3	2	5	2	18	141	60
13	Hose, discharge, 6-in. x 50 ft.	3.3.2.8.2			120		120		370	
14	Pipe, culvert (in feet)	3.3.2.9							120	120
15	Nozzle, servicing, gravity	3.3.2.10.1				6			12	6
16	Nozzle, pressure, 2-inch	3.3.2.10.2	2						4	12
17	Nozzle, pressure, 4-inch	3.3.2.10.2			3				3	
18	Dry-break check valve, 2-inch	3.3.2.11								18
19	Tank chest for fabric tanks	3.3.2.12			2		2	6	36	6
20	General repair kit	3.3.2.13.1							1	1
21	Hose evac. kit, 4-inch	3.3.2.13.2							2	
22	Hose evac. kit, 6-inch	3.3.2.13.2							2	
23	Manifold assy. 4-inch	3.3.2.14						1	5	1
24	Manifold assy. 6-inch	3.3.2.14			1		1		3	
25	Pump and engine assembly	3.3.2.15		1	2		1	1	10	3
26	Cplg. assy. male flg. 3-inch	3.3	1						2	
27	Cplg. assy. male flg. 4-inch	3.3	1						2	
28	Cplg. assy. male flg. 6-inch	3.3	1						2	
29	Cplg. assy. male flg. 8-inch	3.3	1						2	
30	Elbow, hose, Q.D. 2-inch	3.3	2						4	
31	Indicator, sight 2-inch	3.3	2						4	

FIGURE 1. Typical type I AAFS and type II TAFDS fuel systems.

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Fuel components, assemblies, and systems list. - Continued

Item	Components	Spec. ref.	Assemblies[1]						Systems	
			FAA	DUA	BUA	FDA	BPA	TFA	AAFS type I	TAFDS[2] type II
32	Indicator, sight, 6-inch	3.3			1				1	
33	Manifold, dispensing	3.3	1			2			6	6
34	Manifold, drum unloading	3.3		2					2	6
35	Reducer, 6C x 4A	3.3			2		1	1	9	3
36	Reducer, 4C x 6A	3.3			1			1	6	
37	Stand, fuel nozzle	3.3				3			6	12
38	Valve, drum unloading	3.3		6					6	6
39	Valve, gate, 4-inch	3.3			2		2	12	66	20
40	Wye, ACC (adptr-cplr-cplr)	3.3		1				3	16	6
41	Wye, CAA (cplr-adptr-adptr)	3.3			1	1	1	3	20	6

NOTES:

[1] One complete type I (AAFS) consists of the quantity of each assembly component multiplied by the number of assemblies (see legend below), plus components not designated as part of an assembly and additional amounts of selected components for system support/spares.

[2] Quantities in this column make up one complete type II (TAFDS) system.

LEGEND

FAA - Fuel adapting assembly, 2 assemblies required.
DUA - Drum unloading assembly, 1 assembly required.
BUA - Beach unloading assembly, 1 assembly required.
FDA - Fuel dispensing assembly, 2 assemblies required.
BPA - Boost pump assembly, 2 assemblies required.
TFA - Tank farm assembly, 5 assemblies required.

This figure is for information only and is not intended for ordering data.

FIGURE 1. Typical type I AAFS and type II TAFDS fuel systems. - Continued

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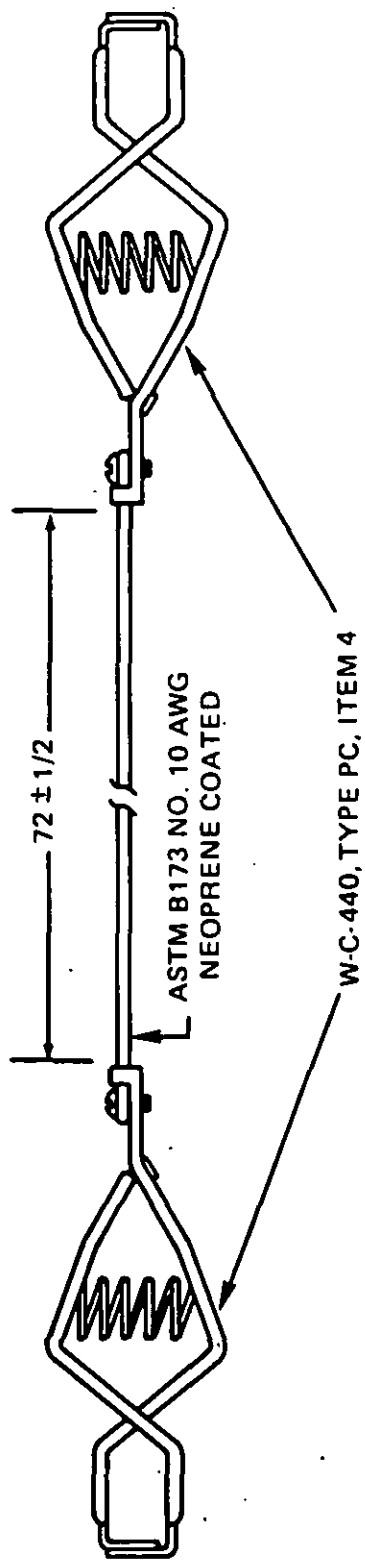


FIGURE 2. Ground wire assembly.

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-82129B(MC) Fuel Systems, Amphibious Assault and Tactical, etc.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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